Expectations:

X: RV $\quad x \in \mathcal{X} \subseteq \mathbb{R}$

discrete — PMF P_X — $E[X] = \sum_{x \in \mathcal{X}} x P_X(x)$

continuous — PDF f_X — $E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$

Bounded support: $\mathcal{X} \to$ bounded $\Rightarrow E(x)$ exists

Unbounded support:

$E[X]$	discrete $\sum_{x \in \mathcal{X}, x>0} x P_X(x)$	discrete $\sum_{x \in \mathcal{X}, x<0} x P_X(x)$	continuous
exists	finite	finite	"
∞	infinity	finite	(instead of summation, we take integral)
$-\infty$	finite	-infinite	
dNE	infinity	-infinity	

$X \sim Geo(p) \quad x \in \{1, 2, \dots, k, \dots\}$

$$E[X] = \sum_{k=1}^{\infty} k(1-p)^{k-1} \cdot p = p \sum_{k=1}^{\infty} k(1-p)^{k-1}$$

$$= p\left[\frac{1}{p} + \frac{(1-p)}{p^2}\right] = 1 + \frac{1-p}{p} = \frac{1}{p}$$

Expectations of functions of RVs

X: RV $\quad g(x) \quad Y = g(x)$

$E[Y] = \sum_{y \in Y} y P_Y(y)$

$E[g(x)]$

LOTUS [Law of the unconscious statisticians]

discrete: $E[g(x)] = \sum_{x \in \mathcal{X}} g(x) P_X(x)$

continuous: $E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$ "existance assumed"

$X_1, X_2, \ldots X_n \sim$ joint distribution

$$E[g(x_1, x_2, \ldots x_n)] = \sum_{x^n} g(x^n)\, p_{x^n}(x^n)$$

$$= \sum_{x^n} g(x_1, \ldots x_n)\, P_{x_1 \ldots x_n}(x_1 \ldots x_n)$$

Properties:

① $E[ax+b] = aE[x]+b$ (doesn't matter if x_i is LI / LD)

② $E[a_1x_1 + a_2x_2 \ldots + a_nx_n + b] = a_1E[x_1] + a_2E[x_2] + \ldots + b$
(for any joint distribution)

Throw a die twice independently (x_1, x_2)

$E[x_1] = 3.5 = E[x_2]$

$E[x_1 + x_2] = 7$

$E[x_1 x_2] = \sum_i \sum_j \frac{ij}{36} = \sum_i \sum_j ij\, P_{x_1x_2}(i,j)$

$= \sum_i \sum_j i P_{x_1}(i)\, j P_{x_2}(j) = \left(\sum_i P_{x_1}(i).i\right)\left(\sum_j P_{x_2}(j)\right)$

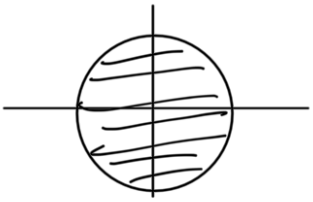
independance is needed

③ if x_1, x_2 independent

$E[g(x_1)\, g(x_2)] = E[g(x_1)]\, E[g(x_2)]$

$x_1, x_2 \sim$ iid $[0,1] \rightarrow$ repeat above

$x_1, x_2 \sim$ uniform on unit circle



$E[x_1] = 0 \quad E[x_2] = 0$

$E[x_1 x_2] = 0$ (symmetry)

$$E[x_1^2 + x_2^2] = \int_{-1}^{1} \int_{-\sqrt{1-x_1^2}}^{\sqrt{1-x_1^2}} (x_1^2 + x_2^2).\frac{1}{\pi}\, dx_2\, dx_1 \quad \text{(or polar)}$$

$X_1, X_2 \sim$ iid $\quad E\left[\frac{X_1}{X_1+X_2}\right] = E\left[\frac{X_2}{X_1+X_2}\right] = \frac{1}{2}$

$X_1, X_2, \ldots, X_n \sim$ iid Bernoulli(p)

$Y = X_1 + X_2 + \ldots \sim$ Bernoulli (n, p)

$Y = \sum_i X_i \Rightarrow E[Y] = np$

$Y^2 = \sum_i X_i \sum_j X_j = \sum_i \sum_j X_i X_j \Rightarrow E[Y^2] = np + n(n-1)p^2$

$= n^2p^2 + np(1-p)$

Variance: characterize the spread about com

$E[X] = 0$

$$\text{var}(X) = E\left[(X - E[X])^2\right]$$

Standard deviation $\sigma(X) = +\sqrt{\text{var}(X)}$

(for $\text{var}(X) = 0$, there should be only 1 pt)

$$\text{Var}(X) = E\left[X^2 - 2E[X]X + E[X]^2\right]$$
$$= E[X^2] - 2(E[X])^2 + (E[X])^2$$
$$= E[X^2] - (E[X])^2 \geq 0$$

$$\Rightarrow E[X^2] \geq (E[X])^2$$

① $X \sim$ Bernoulli (p)

$E(X) = p$

$\text{Var}(X) = p - p^2 = p(1-p)$

maximal when $p = 1/2$

② $X \sim$ Bernoulli (n, p)

$E[X] = np \quad \text{Var}(X) = np(1-p)$

③ $X \sim$ poisson (λ)

$E[X] = \lambda \quad \text{Var}(\lambda) = \lambda$

④ $X \sim$ Uniform (a, b)

$E(X) = \frac{a+b}{2} \quad \text{Var}(X) = \frac{(b-a)^2}{12}$

⑤ $X \sim$ pdf $\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$E[X] = \mu \quad Var(X) = \sigma^2$

Properties:

① $Var(a+X) = Var(X)$

② $Var(aX) = a^2 Var(X)$

③ $X_1, X_2 \sim$ joint distribution

$$Var(X_1+X_2) = E[X_1^2] + E[X_2^2] + 2E[X_1X_2] - (E[X_1])^2 - (E[X_2])^2 - 2E[X_1]E[X_2]$$
$$= Var(X_1) + Var(X_2) + 2(E[X_1X_2] - E[X_1]E[X_2])$$

correlation b/w X_1, X_2

Correlation between X & Y:

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

Covariance — Correlation

X, Y independent $\Rightarrow Cov(X,Y) = 0$

X, Y are "uncorrelated" if $Cov(X,Y) = 0$

Cauchy Schwartz inequality: $(E[UV])^2 \leq E[U^2]E[V^2]$

equality iff $aU + bV = 0$

correlation coefficient $\rho_{XY} = \frac{Cov(X,Y)}{[Var(X)Var(Y)]^{1/2}} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$

$$-1 \leq \rho_{XY} \leq 1$$

Prediction using mean & median:

X: RV Find d that will predict X

$E[|X-d|]$ — mean of absolute deviation — $d = med(X)$ minimizes

$E[(X-d)^2]$ — mean square error — $d = E[X]$ minimizes

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